



Have S&T Innovation and Educational Development in China's Coastal Provinces and Regions Achieved Synchronization? A threshold approach

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Abstract

This article aims to assess the S&T innovation-education integration (STIEI) mechanism's efficacy in the fast-evolving knowledge economy and to provide strategies for boosting the role of technology and knowledge in economic growth. Using data envelopment analysis (DEA) methods to process STIEI input-output panel data from 2006 to 2015, this paper first uses the coastal regions as an example, which are the frontier of China's development, to measure and compare the efficiency of the three systems, including STIEI, S&T innovation, and education. Then, using the panel threshold regression method, we examine the effects of the innovation environment and the foundation of educational activities on economic performance in the coastal, central, and western regions individually, taking into account the reality of regional disparities in China. The results show that first, the STIEI mechanism, which considers S&T innovation and education as a whole, makes significantly more efficient use of human and financial resources. Second, the STIEI mechanism is unable to function at its best and leads to noticeable regional disparities in coastal regions due to the lack of synchronization between S&T innovation and education. Third, the effects of the innovation environment and the basis of educational activities on economic performance vary significantly across various Chinese areas. Our empirical evidence is valuable for local governments and policymakers to create more effective STIEI strategies and planning in developing countries.

Keywords S&T innovation–education integration · Innovation · Education · Data envelopment analysis · Threshold regression · Heterogeneity

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Introduction

The mechanism of S&T innovation–education integration (STIEI) originates from the ideological principle of ‘unification of teaching and research’ proposed by the German educator Wilhelm Humboldt in the early nineteenth century (Weimer, 1997). STIEI can define as the promotion of the mutual transformation of innovation and educational development outcomes through a close association of innovation and education to achieve a win-win sharing of both, obtaining the highest input-output ratio of both (Linton & Xu, 2020). Since the strategy of “revitalizing the country through STIEI” was proposed in 1995, China has focused on S&T innovation as the primary productive force, adhered to education as foundational, and continuously raised the nation’s awareness of S&T innovation–education (Zhang, 2014). The long-term strategy of “rejuvenating the country through STIEI” has contributed to the construction of *think tanks* and an innovative country under socialism with Chinese characteristics in the new era (Argyropoulou et al., 2019; Huang et al., 2010).

In the rapidly changing knowledge economy, to seize the opportunities and initiatives of innovative development, developed countries are rushing to implement the S&T innovation-driven development strategy (Khorsheed, 2017). In the era of the knowledge economy (Brostrm et al., 2020), S&T innovation faces a new task of combining original innovation and cutting-edge breakthroughs with the integration of S&T innovation and educational development (Prokop et al., 2021). China is currently in economic transformation and urgently needs to improve the content of technology and knowledge in its economic growth to improve its ability to participate in global economic cooperation and competition. How to further enhance the level of STIEI has become a top priority. Since the reforms and opening up, coastal regions have gradually become the frontier of innovation and educational development in China because of their advantages in location, environment, and policies (Shi et al., 2022; Wu et al., 2019). Therefore, the level of STIEI in coastal regions can reflect the feasibility of implementing STIEI mechanisms in China. However, due to the significant regional differences within coastal regions (Wu et al., 2019), the actual S&T innovation and educational development may not be synchronized, which can hinder the effectiveness of the STIEI mechanism. Therefore, in the context of the knowledge economy and the new development pattern, there is an urgent need to evaluate the feasibility of the STIEI mechanism in China and find the path to increase the content of technology and knowledge in economic growth.

This study aims to provide answers to the following two questions, which are our contributions. First, how to verify the feasibility of the STIEI mechanism in China? To the best of our knowledge, comparing the efficiency of innovation and the efficiency of education can see their respective development, but it is conducive to determining if their integration is advantageous (Bonaccorsi et al., 2022; Guccio et al., 2016). From the system perspective (Colapinto & Porlezza, 2012), we constructed the input–output analysis framework of STIEI. Taking the coastal region of China as an example, we measured and compared the efficiency of three

systems, including STIEI, S&T innovation, and education, to verify the feasibility of implementing the STIEI mechanism in China. Our work not only complements the existing literature but also enriches the STIEI theory. Second, how to improve the content of technology and knowledge in economic growth to further enhance STIEI? Broadly, technological progress refers to the accumulation and improvement of various forms of knowledge content covered by technology. The desired results of the STIEI mechanism are also equal to technological progress in a broad sense. The actual value brought by such technological progress can measure by economic performance. Using economic performance as the desired results from STIEI, we investigated the impact of innovation environment and the foundation of educational activities on economic performance in different regions of China, intending to provide theoretical and practical references for promoting the level of STIEI in the era of the knowledge economy.

The remainder of the article is organized as follows: the “Literature Review” section is a review of related work; the “Methodology and Data” section includes the theoretical basis for analyzing the relationship between the three systems of STIEI, S&T innovation and education, the methods and data used in the study; the “Empirical Results” section discusses the characteristics of efficiency, and the influence of innovative environment and educational development base on economic performance in different regions; the “Conclusions” section summarizes the conclusions of the paper and proposes policy recommendations.

Literature review

By reviewing the existing literature, we found that after the originating idea of STIEI, “the unification of research and teaching” was proposed, it has triggered extensive discussions among scholars on the issue of the relationship between research and teaching. However, these discussions have never formed a consensus (Burke-Smalley et al., 2017; Cao et al., 2021; Hattie & Marsh, 1996). Compared with the ‘theory leads practice’ evolutionary path of STIEI in developed countries, the development of STIEI theory in China significantly lags behind practice, but both domestically and internationally, STIEI faces a prominent imbalance in the development process (Huang, 2018). Taking universities as an example, Huang (2018) found that innovation outputs are more visible and quantifiable than education outputs and are more likely to be the baton for resource allocation and performance assessment in universities. In this way, the relationship between research and teaching is always unbalanced in their integration. To improve the level of STIEI, the S&T innovation and education input–output profiles in the system need to rationalize to enhance the efficiency of their resource allocation. Most relevant studies have focused on separate and comparative evaluations of innovation efficiency and education efficiency, and there is a relative lack of research on the efficiency of STIEI (Bonaccorsi et al., 2022; Guccio et al., 2016; Huang et al., 2021; Shi et al., 2022). Next, we focus on sorting out the relevant studies on S&T innovation and education efficiency.

Research perspectives for S&T innovation mainly focus on international and provincial (municipal or industry, universities, and enterprises) contexts, and its content

involves S&T innovation input–output models, innovation network construction, and innovation impact mechanisms (Barra & Zotti, 2018; Guccio et al., 2016; Shi et al., 2022; Wu et al., 2019). Wang et al. (2019) analyzed regional differences in the level of innovative efficiency in China and found that the eastern region leads other regions. Zhu et al. (2022) have found that improving technological efficiency can help strengthen the productivity of S&T innovation. Other studies have found that the spillover of intellectual capital is an important factor affecting the efficiency of national S&T innovation (Ramesh, 2013). Technological progress and increased investment in S&T research have significantly helped improve innovative efficiency (Fritsch & Slavtchev, 2011; Xia et al., 2019). Also, the characteristics and differences of the time series of innovative efficiency between regions are significant and influenced by national policies; this is also true of enterprises, universities, and industries (Barra & Zotti, 2018; Fei et al., 2014).

The existing literature has a variety of approaches to studying the efficiency of educational input–output, including mainly parametric and non-parametric methods (De Witte & Lopez-Torres, 2017; Huang et al., 2021; Lafuente et al., 2016). The non-parametric method has been widely used by scholars to evaluate the public sector's input–output efficiency (Kauppinen, 2016; Tone & Tsutsui, 2014). Education is one of the first fields to use this method because of its comprehensive advantage (Färe et al., 1992). Most educational input–output research focuses on the distribution of educational resources, the evaluation of educational input–output efficiency (internal and external), the decomposition index of total factor productivity (TFP), and the analysis of regional differences in education performance among different universities (Abbott & Doucouliagos, 2003; Agasisti et al., 2020; Huang, 2018; Huang et al., 2021).

Overall, the well-established experience of the above literature provides the basis for this paper's analysis of the theoretical relationship between the three systems of STIEI, S&T innovation, and education based on system theory and resource flow theory (Deng, 1989; Murgia, 2018; Esther et al., 2019). In addition, it provides a methodological reference for the empirical analysis of this paper, mainly including the evaluation of regional efficiency and the research on finding the path to increase the technological and knowledgeable content in economic growth.

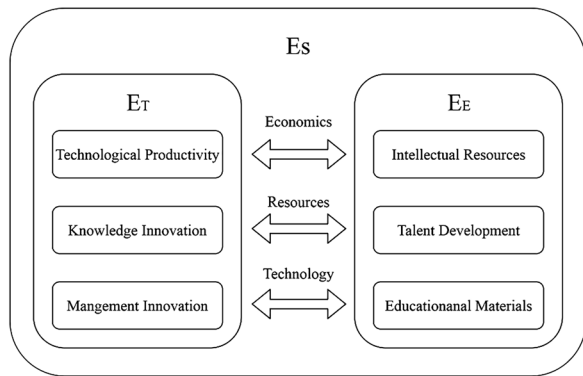
Methodology and Data

Theoretical Basis

The efficiency of STIEI is the performance of the output obtained by the unit's input of S&T innovation–education resources based on ensuring the environment and resource allocation. It can reflect the ability to operate, transform, produce, and allocate S&T innovation–education resources and the input–output level of the combined development of S&T innovation and education (Agasisti et al., 2012).

According to system theory (Colapinto & Porlezza, 2012), the STIEI input–output system includes a combination of two subsystems: S&T innovation input–output and educational input–output. The elements in each system can function effectively together

Fig. 1 Schematic diagram of development across the three systems



because of the intersection of the development links, more effectively than if the system had developed individually. This input–output system coordinates and integrates the resources and economy in the two subsystems, thereby promoting the coordinated development of the input and output factors between the systems (Murgia, 2018). The functional relationship between the three systems is $E_S = (E_T, E_E, \delta)$, where E_S refers to the input–output system of STIEI, E_T refers to the input–output subsystem of S&T innovation, and E_E refers to the input–output subsystem of education, δ represents the collection of relationships between the subsystems.

According to the theory of resource flow, element flow and exchange allocation may occur between systems (Esther et al., 2019) because of the differences in resource endowments, economic foundations, and human management experience. E_T continuously embodies the practicality of technology, productivity, and S&T achievements, while the demand for technological talents increases. Further, E_E can provide a development environment for talent training with knowledge input and quality improvement. The level of E_E determines the level of development in the high-tech field, and sustainable development. However, E_T can support economic development and provide intellectual resources for economic development. As the two subsystems complement each other, they can achieve advantages in economic production. A schematic diagram of the development across the three systems is shown in Fig. 1.

Methodology

Data Envelopment Analysis

Data envelopment analysis (DEA) is a commonly used efficiency measurement method that constructs a non-parametric envelope frontier surface from the spatial panel data of the decision unit (Kirby et al., 2019). All frontier surfaces belong to valid points, while those outside the frontier surface belong to invalid points (Kauppinen, 2016). The traditional CCR (A. Charnes, W. W. Cooper, and E. Rhodes) model does not consider the input and output relaxation variables that affect the reliability of the measured efficiency value. Tone (2002) added relaxation variables to the proposed non-radial model to maximize the actual profit of the decision-making

unit (DMU). To overcome a situation in which multiple DMUs may be effective, Tone and Tsutsui (2014) proposed a super-efficiency slacks-based measure (SBM) model, the basic principle of which involves ranking and measuring effective DMUs when evaluating DMUs. For the invalid unit, the evaluation result is still consistent with the SBM model because the effective frontier has not changed.

To facilitate the ranking and analysis of effective DMUs, the study attempts to use the super-efficiency SBM model to measure the efficiency of STIEI. The model is as follows:

$$\rho^* = \frac{\frac{1}{m} \sum_{i=1}^m \frac{s_i^-}{x_{i0}}}{\frac{1}{s_1} \sum_{r=1}^{s_1} \frac{s_r^g}{y_{r0}}} \quad s.t. \begin{cases} x_0 = X\lambda + s^- \\ y_0^g = Y^g\lambda - s^g \\ s^- \geq 0, s^g \geq 0, \lambda \geq 0 \end{cases} \quad (1)$$

where ρ^* is the calculation of efficiency values; m and s_1 are the number of input and expected output factors, respectively; i and r represent the year of the research period; and $s = (s^-, s^g)$ represent the input and expected output slack; x and y^g are the input and expected output values, respectively; λ is the weight vector; 0 in the model is the evaluated unit; X and Y^g are the matrices of input and expected output, respectively. Still, this method by formation is argued by the related literature to measure efficiency value as too low. Accordingly, we also used the conventional quantity-based (Malmquist index) measures of efficiency (for radial estimation) and compared them with the results of slacks-based measures for robustness (Alimohammadi & Mohammadi, 2016; Malmquist, 1953; Spitsin et al., 2022).

Threshold Regression Model

Due to differences in economic aggregates, technological talent advantages, policy environment, and educational environment among China's coastal regions, the differences in economic performance (EP) between regions have fluctuated and shrunk during the study period. This leads us to ponder that the factors affecting the economic performance of the coastal regions may not be linear. Therefore, we use the panel threshold regression model proposed by Hansen (1999) to analyze and discuss the factors that affect EP. The model is determined by the endogenous decision of the sample data to determine the threshold, so it has strong objectivity and uses bootstrap methods to estimate the accuracy of the threshold (Huang et al., 2012). The idea of single-threshold regression is that when there is a threshold for a certain variable A , when $A > a$ or $A \leq a$, the influence of the variable on the dependent variable will be significantly different. The set model is as follows:

$$Y_{it} = \alpha X_{it} + \xi_1 K_{it} \times I(K_{it} \leq \gamma_1) + \xi_2 K_{it} \times I(\gamma_1 < K_{it}) + C + \varepsilon_{it} \quad (2)$$

where i is the research area; t is the time; Y indicates the dependent variable; K indicates the threshold variable; X refers to a set of variables; γ represents the threshold value; α is the coefficient of influence of X on the dependent variable; ξ_1 and ξ_2 respectively represent the coefficient of influence of the threshold variable K on the dependent variable Y when the threshold variable $K_{it} \leq \gamma_1$, $K_{it} > \gamma_1$; C is a constant

coefficient; ε_{it} is a random disturbance term; $I(\cdot)$ is a qualitative function. The dual threshold model is set up as follows:

$$Y_{it} = \alpha X_{it} + \xi_1 K_{it} \times I(K_{it} \leq \gamma_1) + \xi_2 K_{it} \times I(\gamma_1 < K_{it} \leq \gamma_2) + \xi_3 K_{it} \times I(K_{it} > \gamma_2) + C + \varepsilon_{it} \quad (3)$$

We first need to minimize the residual sum of squares of γ and $S(\gamma)$ to obtain the estimated value of γ and then obtain the estimated values of other parameters, and finally perform two tests: The first is to check the effect of the threshold, that is, to check the existence of the threshold and the number of thresholds. Let the original hypothesis be $H_0: \xi_1 = \xi_2$, the corresponding alternative hypothesis is $H_1: \xi_1 \neq \xi_2$, and use the F statistic and σ^2 statistic test of the residual. The second is to test the threshold value. The original hypothesis is that the estimated threshold value is equal to the true value. The maximum likelihood statistic $LR(\gamma)$ test was used (Al-Hassan, 2013).

According to models (2) and (3), we describe the threshold regression model designed in this paper.

$$\begin{aligned} EP_{it} = & \mu_1 X_{it} + \rho_1 EFRS \times I(EFRS \leq \sigma_1) + \rho_2 EFRS \\ & \times I(\sigma_1 < EFRS \leq \sigma_2) + \rho_3 EFRS \\ & \times I(EFRS > \sigma_2) + C_1 + \varepsilon_{it1} \end{aligned} \quad (4)$$

$$\begin{aligned} EP_{it} = & \mu_2 X_{it} + \varphi_1 STMS \times I(STMS \leq \xi_1) + \varphi_2 STMS \\ & \times I(\xi_1 < STMS \leq \xi_2) + \varphi_3 STMS \\ & \times I(STMS > \xi_2) + C_2 + \varepsilon_{it2} \end{aligned} \quad (5)$$

$$\begin{aligned} EP_{it} = & \mu_3 X_{it} + \omega_1 STML \times I(STML \leq \pi_1) + \omega_2 STML \\ & \times I(\pi_1 < STML \leq \pi_2) + \omega_3 STML \\ & \times I(STML > \pi_2) + C_3 + \varepsilon_{it3} \end{aligned} \quad (6)$$

where EP is the dependent variable. X is a group of control variables that includes local students scale (LSS), R&D intensity (RDI), and teaching staff construction (TSC); μ_n is the influence coefficients of the control variables; ρ_k , φ_k , ω_k are the influence coefficients of the threshold variables that includes educational financial resources support (EFRS), S&T market scale (STMS), and S&T manpower level (STML); σ_m , ξ_m , π_m are the double thresholds of each threshold variable; C_n and ε_{itm} are the constant term and random disturbance term of the regression equation, respectively ($n = 1, 2, 3$; $k = 1, 2, 3$; $m = 1, 2$).

Data

The selection of indicators is based on the principles of scientificity, indicator representativeness, and data availability (Balland & Boschma, 2021). This paper first takes the coastal regions (including Liaoning, Tianjin, Hebei, Shandong, Jiangsu, Zhejiang, Shanghai, Fujian, Guangdong, Guangxi, and Hainan), which are the

frontier of China's development, as an example, to measure and compare the efficiency of three systems, including STIEI, S&T innovation, and education. Then, to enhance the generalizability of the results, the impact of the innovation environment and the foundation of educational activities on economic performance in different regions of China is investigated. This study uses panel data from 30 provinces (excluding Hong Kong, Macao, Taiwan, and Tibet) in China from 2006 to 2015.¹

Efficiency Measurement Input–Output Indicators

The definition of the input and output indicators used in the study is given in Table 1. It is important to note that the full stage of education includes higher education, high school, vocational school, junior high school, and elementary school. In addition, considering the time lag of paper publication, the indicator of the number of papers purchased by three major institutions is selected with a two-period lag. We processed the input and output indicators using the entropy method to obtain two inputs and two outputs, ensuring that the sum of input and output variables is not less than one-third of the number of DMUs (Spitsin et al., 2022).

Threshold Regression Model Indicators

The selection of indicators for threshold regression is shown in Table 2. The dependent variable is economic performance (EP). We use EP to characterize the desired results from STIEI, measured through economic total factor productivity. It is measured through the Malmquist index method (Spitsin et al., 2022), where the input indicators include the total investment in fixed assets and the number of employed people in the society, and the output indicator is GDP. Moreover, related price data from each year were flattened to a constant price value using 2006 as the base year, using the corresponding price index.

The threshold variables mainly include the S&T innovation environment and the foundation of educational activities. S&T innovation environment can define in terms of manpower level and market scale. S&T manpower level (STML) is measured by the full-time equivalent of R&D personnel which can reflect investment in human resources in innovative activities (Ma & Zhou, 2012). S&T market scale (STMS) is measured by the contract value of the technology market (Pan et al., 2020). The foundation of educational activities can define in terms of educational financial resources support (EFRS), which is measured by the public financial education funds (Huang et al., 2021).

Control variables are as follows: R&D investment can define in terms of R&D intensity (RDI), which is measured by the internal expenditure of R&D (Qin & Du, 2018). Faculty strength can be defined in terms of teaching staff construction (TSC),

¹ All data from 2007 to 2018 were retrieved from China Statistical Yearbook, China Science and Technology Statistical Yearbook, China High-tech Statistical Yearbook, China Educational Expenditure Statistical Yearbook and China's coastal provinces and regions statistical yearbooks and statistical bulletins. Based on the National Bureau of Statistics, the central region includes Shanxi, Heilongjiang, Jilin, Anhui, Jiangxi, Henan, Hubei, and Hunan; and the western region includes Inner Mongolia, Chongqing, Sichuan, Guizhou, Yunnan, Ningxia, Shaanxi, Gansu, Qinghai, and Xinjiang.

Table 1 STIEI input and output indicators of DEA model

Primary indicators	Secondary indicators	Content of indicators	References
STIEI input	Innovation human resource input	The full-time equivalent to R&D personnel	Ma and Zhou (2012)
	Education human resource input	The number of full-time teachers (including all the stages of education)	Huang et al. (2021)
	Innovation financial resources input	The internal expenditure on R&D	Qin and Du (2018)
	Education financial resources input	The public financial education funds	Huang et al. (2021)
	Innovation output	The number of papers purchased by three major institutions (science citation index, engineering index, conference proceedings citation index–science)	Sharma and Thomas (2008); Wang et al. (2019);
STIEI output		The number of domestic patent applications (invention, utility model, design)	
	Education output	The contract value of the technology market The number of students (including all the stages of education)	Hou et al. (2019)

Table 2 The selection of indicators for the threshold regression

Indicators	Indicator abbrevia- tions	Content of indicators
Economic performance	EP	Economic total factor productivity
S&T manpower level	STML	The full-time equivalent of R&D personnel
S&T market scale	STMS	The contract value of the technology market
Educational financial resources support	EFRS	The public financial education funds
R&D investment	RDI	The internal expenditure of R&D
Faculty strength	TSC	The number of full-time teachers at all stages after stratified weighted summation
Talent training scale	LSS	The number of students at all stages after stratified weighted summation

Table 3 Value of efficiency of STIEI in China's coastal regions

Region	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
Fujian	0.9316	0.9292	0.9199	0.9329	0.9558	0.9453	0.8518	0.9287	0.9170	0.8171
Guangdong	1.0040	1.0073	1.0099	1.0124	1.0165	0.7274	1.0080	1.0080	0.8632	0.7385
Guangxi	1.0225	1.0216	1.0140	1.0194	1.0227	1.0081	1.0336	1.0389	1.0477	1.0097
Hainan	0.9761	0.9762	0.9740	0.9783	0.9788	0.9875	0.9782	0.9807	0.9754	0.8843
Hebei	0.9596	1.0010	1.0060	1.0054	1.0027	1.0301	0.9540	0.9544	0.9448	0.8419
Jiangsu	0.8794	0.8773	0.8601	0.8647	0.8754	0.9095	1.0277	0.9127	0.8861	0.7762
Liaoning	0.9328	0.9434	0.9471	0.9590	0.9612	0.9985	1.0009	0.9769	1.0038	1.0104
Shandong	0.8573	0.8553	0.8254	0.8395	0.8161	0.8521	0.8396	0.8506	0.8426	1.2512
Shanghai	1.2020	1.1946	1.1622	1.1711	1.1292	1.0908	1.0650	1.0764	1.0752	1.0893
Tianjin	0.9377	0.9322	0.9243	0.9419	0.9392	0.9521	0.9378	0.9529	0.9818	0.8982
Zhejiang	0.8996	0.8855	0.9061	0.8995	0.8848	0.9205	0.8881	0.9314	0.9011	0.7821
Mean	0.9642	0.9659	0.9591	0.9658	0.9621	0.9479	0.9617	0.9664	0.9497	0.9183

which is measured by the number of full-time teachers at all stages after stratified weighted summation (Hou et al., 2019). The talent training scale can define in terms of the local students scale (LSS), which is measured by the number of students at all stages after stratified weighted summation (Huang et al., 2021).

Empirical results

Evolution of Efficiency in Time and Space

The study uses MaxDEA 8.0 software to process the STIEI input–output panel data in China's coastal regions and calculates the efficiency of STIEI, as shown in

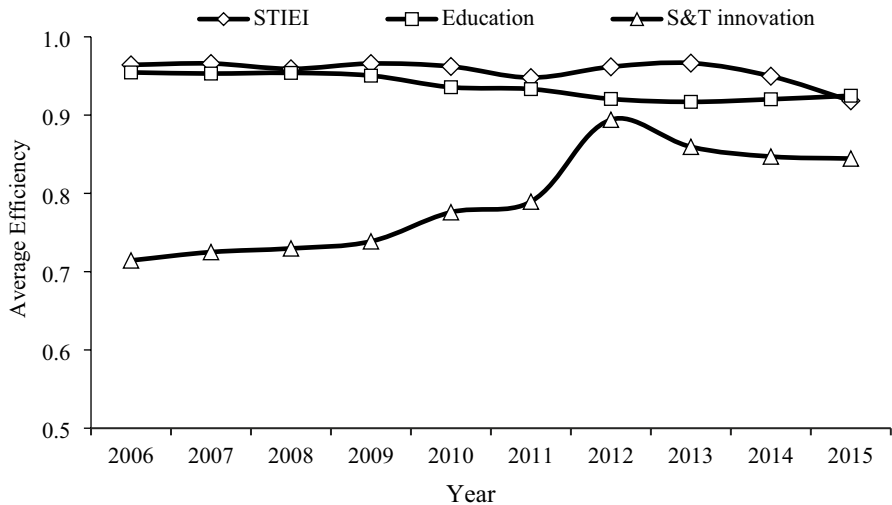


Fig. 2 Average efficiency evolution of the three systems from 2006 to 2015 in China's coastal regions

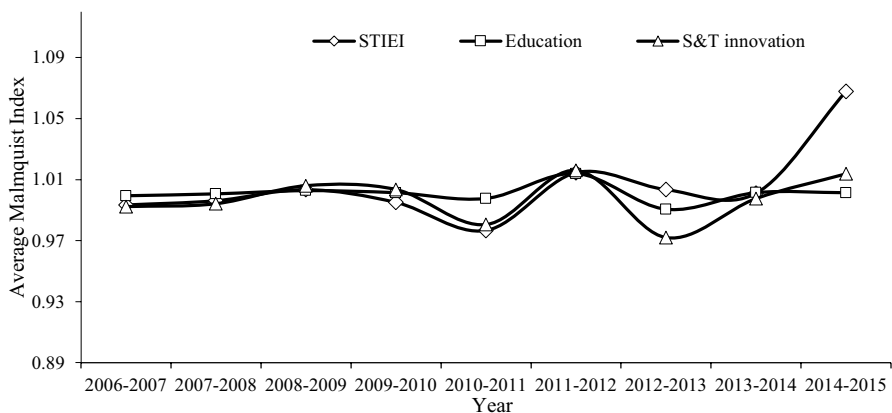


Fig. 3 Average Malmquist index evolution of the three systems from 2006 to 2015 in China's coastal regions

Table 3. The average efficiency evolution of the three systems in time is visualized in Fig. 2. The average Malmquist index evolution of three systems in time is visualized in Fig. 3.

Temporal Evolution of Efficiency

As shown in Table 3 and Fig. 2, the average efficiency of STIEI was 0.9561, which was higher than the average efficiency of S&T innovation (0.7918) and education (0.9363). To enhance the robustness of our findings, we proceeded to compare the average Malmquist index evolution of the three systems in Fig. 3. We found that the

average index of STIEI (1.0057) remained the highest during the study period, compared to S&T innovation (0.9973) and education (1.0010). This result shows that STIEI has relatively higher efficiencies in the use of financial and human resources and validates the feasibility of this mechanism in China. However, the trends of integration and educational efficiency tended to be consistent. There could be a lack of synchronization between S&T innovation and education development.

The average efficiency of STIEI in coastal regions fluctuated by 4.76%, and the fluctuations during the 12th Five-Year Plan are greater than those in the 11th Five-Year Plan. In 2006, the average efficiency was 0.9642. In the following years, it fluctuated within a small range until it decreased slightly to 0.9621 in 2010. Overall, during the 11th Five-Year Plan, the average efficiency fluctuation dropped slightly by 0.211%. During the 12th Five-Year Plan, the average efficiency fluctuation dropped by 3.13%. The main reason is the relative imbalance between the process of S&T innovation (E_T) and the effectiveness of education (E_E). Because E_E has a more developed infrastructure, policies, and regulations than E_T (Vallance, 2016), the efficiency gap between E_E and E_T was more pronounced in the early stages of the study.

As education outputs see tangible results, E_E can provide support and input resources for E_T . In July 2011, the National 12th Five-Year Plan for S&T development proposed the strategic plan ‘Innovation 2020’, which has significantly reduced the difference in efficiency levels between the two subsystems. Policy-driven improvements in E_T can also be effective in improving E_E because S&T innovation can penetrate all aspects of education (Dubina et al., 2012). The efficiency of E_s increases when both E_E and E_T achieve complementary strengths. As the efficiency gap between E_E and E_T narrowed, the development of STIEI gradually stabilized. However, strengthening human and financial resources for STIEI in the short term will cause E_E and E_T to be out of sync. Therefore, at the end of the study, there is a separation in the pace of E_E and E_T .

Evolution of the Spatial Pattern of Efficiency

ArcGIS was used to characterize the spatial pattern of efficiency. The initial and final years of the 11th Five-Year Plan and the 12th Five-Year Plan (2006, 2010, 2011, 2015) were selected as the spatial pattern evolution of the STIEI efficiency in China’s coastal regions (Fig. 4).

The potential of STIEI in the northern coastal regions began to emerge during the 11th Five-Year Plan. In 2006, among the 11 coastal provinces, the integrated efficiency of Shanghai (1.202), Guangdong (1.004), and Guangxi (1.023) were all greater than 1, while those of the remaining provinces were less than 1. The integration efficiency exhibited an overall upward trend in 2010 compared to 2006, and the difference in efficiency between regions is shrinking. The efficiencies of Hebei (1.003), Shanghai (1.129), Guangdong (1.017), and Guangxi (1.023) were all greater than 1 in 2010, while those of remaining areas were less than 1. The spatial pattern of efficiency showed that the efficiencies of the northern, central, and southern provinces were completely effective, and the north-central and central-southern provinces had efficiency losses.

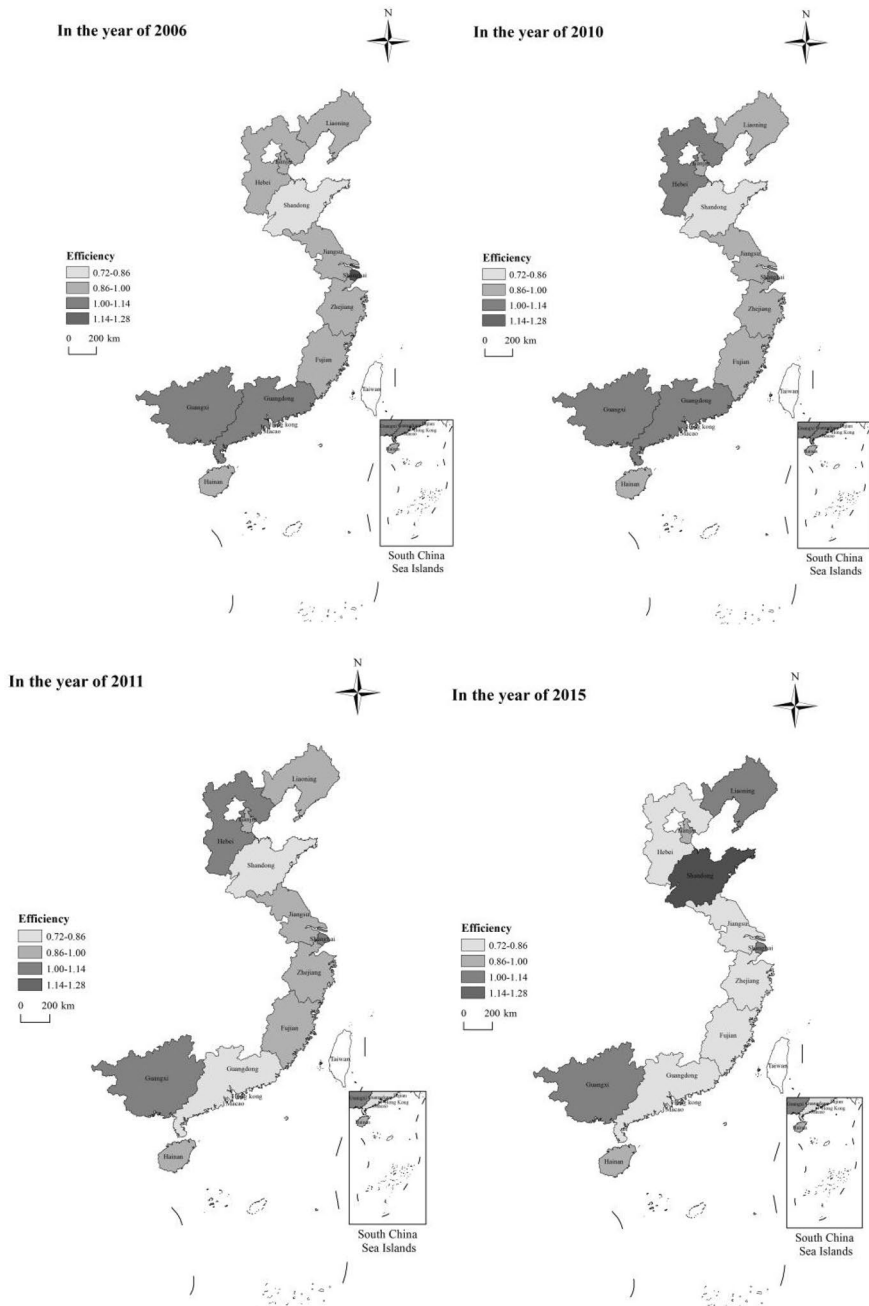


Fig. 4 Spatial pattern of the efficiency of STIEI in China's coastal regions

There were significant regional differences in integrated efficiency during the 12th Five-Year Plan. In 2011, only three provinces had efficiencies above 1, namely Hebei (1.030), Shanghai (1.091), and Guangxi (1.008). Compared with efficiencies of less than 1 in 2010, the efficiencies of the northern-central provinces have shown an overall upward trend, while those of some southern-central provinces have declined, and regional differences have gradually widened. Overall, the efficiencies of the northern-central provinces were higher than that of the southern-central provinces. As of 2015, integrated efficiency levels showed a downward trend compared with 2011, and the efficiency difference between regions was significant. The integrated efficiency of Shanghai (1.090), Shandong (1.253), Liaoning (1.010), and Guangxi (1.010) are effective, while the efficiencies of other provinces have declined. The picture presents a three-pronged efficiency development pattern of north, central, and south.

During the study period, two aspects affected changes in the spatial pattern of integrated efficiency. First, the training of educational talents lacked full-stage technological innovation empowerment. China's coastal regions with faster economic growth have significant advantages in the innovative environment and systems. For example, five provinces, namely Tianjin, Jiangsu, Shanghai, Zhejiang, and Guangdong, have invested more than 100 billion yuan in R&D expenditure, accounting for about 50% of the country overall. However, funding in some provinces is mainly focused on the higher education level, leading to an imbalanced development between innovation and other stages of education. Some provinces such as Shandong, Liaoning, and Guangxi have made improvements, which emphasized in their 12th Five-Year Plan that R&D development should focus on providing support for upgrades and innovation in different educational links.

Second, there is a lack of synchronization between S&T innovation and education. After the strategic plan "Innovation 2020" proposed in 2011, coastal provinces have significantly increased their investment in human and financial resources for S&T innovation, resulting in a significant increase in innovation outputs. However, there is a lag in the translation of innovation outputs into economic benefits and their impact on all elements of education, as well as in the improvement of the innovation environment through reforms in all segments of education. This not only hinders the effectiveness of the STIEI system and leads to a decline in the overall efficiency of STIEI, but also makes regional differences more pronounced.

Threshold Regression Results

Threshold Effect Test and Threshold Value Estimation

We conducted regression analysis in coastal areas with the help of STATA 17.0 software. First, the test results of the threshold effect are shown in Table 4. It can see that the single-threshold effect test of the S&T market scale (STMS), S&T manpower level (STML), and education financial resources support (EFRS) all pass the 5% significance level, and the double-threshold effect test all passed the 1%

Table 4 Test on threshold effects

Index	STMS	STML	EFRS
Single-threshold effect test	22.777*** [0.006]	16.390** [0.026]	11.498*** [0.000]
Double-threshold effect test	17.846** [0.034]	20.836*** [0.000]	8.216*** [0.002]
Triple-threshold effect test	0.000 [0.326]	0.000 [0.330]	0.000 [0.204]
Sample maximum	5.925	50.686	20.220
Sample minimum	0.006	0.121	0.392
Sample mean	1.395	14.174	6.065

The numbers above the square brackets are the F statistics corresponding to the threshold test; ***, **, and * indicate the significance levels at 1%, 5%, and 10%, respectively. The square brackets are the *P* values obtained by repeated sampling using the bootstrap method

Table 5 Estimation of threshold values

Threshold variable	First threshold		Second threshold	
	Estimated value	95% confidence interval	Estimated value	95% confidence interval
STMS	0.037	[0.036, 0.193]	1.007	[0.940, 3.334]
STML	15.336	[4.759, 49.880]	34.469	[34.277, 49.880]
EFRS	4.856	[2.058, 7.958]	13.111	[3.256, 18.506]

significance level, but none passed the triple-threshold effect test, so all four have double thresholds.

Second, the estimated values of the various thresholds are tested to obtain the estimated results of the thresholds in Table 5.

Then, the double threshold map of STMS, STML, EFRS in Fig. 5.

Finally, the estimated values of various parameters are sorted out as shown in Table 6.

Results Analysis

- (1) The impact of EFRS on EP is characterized by a significant positive double threshold in coastal regions. When EFRS is less than 48.56 billion yuan, its elasticity coefficient for EP is 0.068. When this threshold is crossed, the coefficient drops to 0.046. When EFRS exceeds 131.11 billion yuan, the coefficient becomes 0.042. Given the significant regional differences in coastal economic development, the scale, source, and structure of educational resource allocation varies

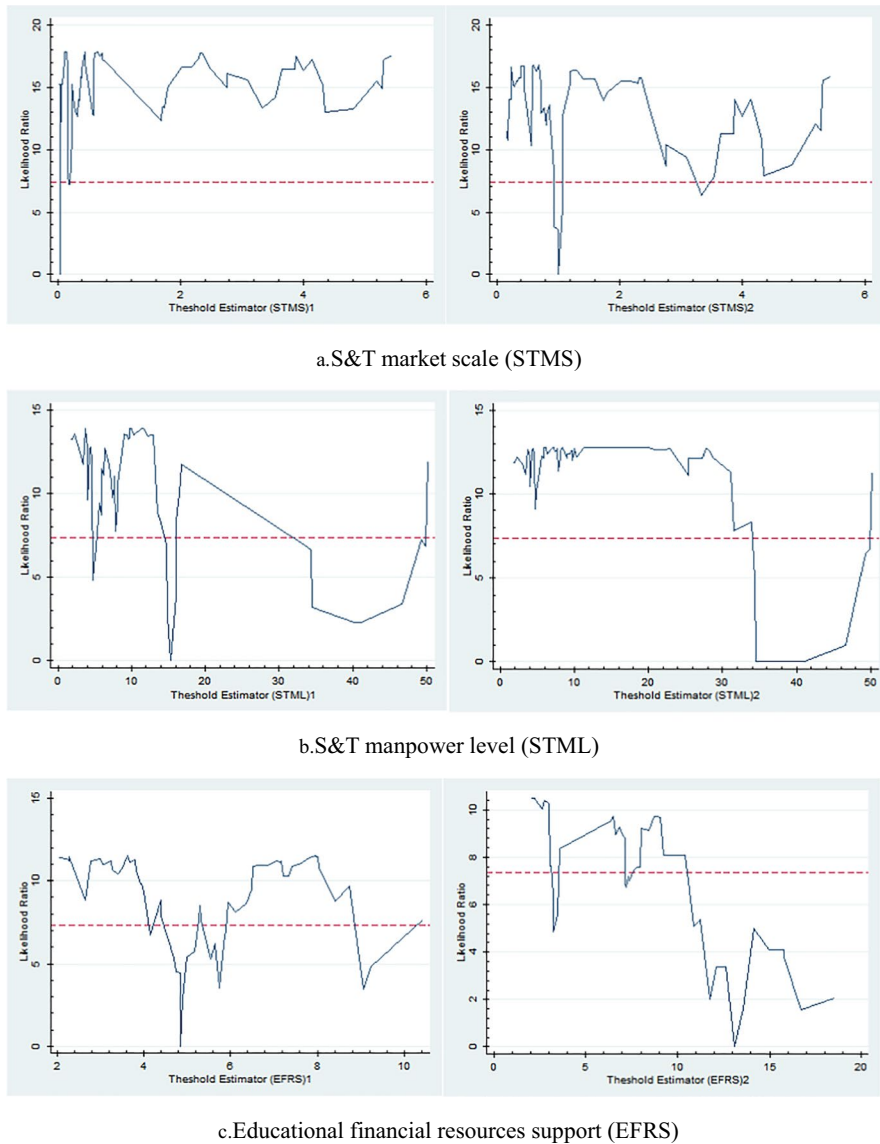


Fig. 5 Double threshold map of S&T market scale (STMS), S&T manpower level (STML), educational financial resources support (EFRS)

from place to place, which can alter the extent to which the EFRS contributes to EP. Before crossing the first threshold of EFRS, with the continuous increase of regional education financial investment, the degree of optimal allocation of education resources will continue to increase, and the promotion of economic effects is obvious. When it is between the double thresholds of EFRS, the intensity of education financial investment will limit by the regional education development

Table 6 Estimation results of model parameters

Variable	Model 1	Variable	Model 2	Variable	Model 3
	Coefficient		Coefficient		Coefficient
TSC	0.670*** (0.165)	TSC	0.711*** (0.167)	TSC	0.806*** (0.156)
LSS	− 1.477*** (0.169)	EFRS	0.024*** (0.009)	EFRS	0.031*** (0.008)
STML	0.004*** (0.00168)	LSS	− 1.262*** (0.170)	LSS	− 1.363*** (0.160)
RDI	− 0.043*** (0.013)	RDI	− 0.029** (0.013)	STML	0.006*** (0.002)
STMS	− 0.016 (0.011)	STMS	− 0.020* (0.011)	RDI	− 0.034*** (0.012)
EFRS_1	0.068*** (0.012)	STML_1	0.015*** (0.004)	STMS_1	− 6.302*** (1.112)
EFRS_2	0.046*** (0.009)	STML_2	0.007*** (0.002)	STMS_2	− 0.170*** (0.033)
EFRS_3	0.042*** (0.009)	STML_3	0.006*** (0.002)	STMS_3	− 0.038*** (0.010)
Constant	1.198*** (0.033)	Constant	1.110*** (0.031)	Constant	1.209*** (0.027)
<i>R</i> -squared	0.845	<i>R</i> -squared	0.735	<i>R</i> -squared	0.754

The *t* values corresponding to each coefficient are in parentheses; ***, **, and * indicate the significance level at 1%, 5%, and 10%, respectively

strength, and redundant investment is prone to appear, which makes the positive promotion of EFRS to EP weakened.

The provinces between the two thresholds of EFRS have the largest number, accounting for 64%, including Tianjin, Liaoning, Fujian, Guangxi, Shanghai, Hebei, and Zhejiang. Observing the per capita financial budget for education of these provinces at the end of the 12th Five-Year Plan period, we found that there are significant differences in the structure of investment in education funding across regions. The provinces that crossed the second threshold were Shandong, Jiangsu, and Guangdong. At the end of the Twelfth Five-Year Plan period, the EFRS scale of these three provinces ranked among the top three coastal provinces. While increasing the EFRS, the education environment and institutional development should further improve to promote the realization of the EFRS scale effect. The province that did not cross the first threshold was Hainan. Because the optimization space of its EFRS is more significant, its EFRS can be further increased.

- (2) The impact of STML on EP is characterized by a significant positive double threshold. When STML is less than 153,360 man-years, its elasticity coefficient

on EP is 0.015. When this threshold is crossed, the coefficient decreases to 0.007. When STML exceeds 344,690 man-years, the coefficient becomes 0.006. The effect of STML investment cannot achieve overnight. It requires the continuous accumulation of long-term S&T human capital. Before crossing the first threshold of STML, with the accumulation of human capital in S&T innovation, the coastal provinces with relatively low economic development, driven by S&T development policies, have significantly increased their rate of technological progress. When it is between the double thresholds of STML, although the total amount of human resources in science and technology has reached a certain height, it is restricted by the structure of S&T talents. When crossing the second threshold of STML, STML gradually realizes the transformation from quantitative change to qualitative change, and the digestion and absorption capacity of new technologies can further enhance, thus stabilizing the role of STML for EP.

The number of provinces that did not cross the first threshold of STML was the largest, accounting for 55%, including Hebei, Tianjin, Fujian, Liaoning, Guangxi, and Hainan. These provinces can further increase innovation investment, and manpower investment, clarify the direction of S&T innovation and guide economic development to improve the efficiency of factor production. The provinces between the two thresholds, including Shanghai, Shandong, and Zhejiang face problems such as the loss of S&T talents and the lack of high-level talents. They should consider the actual situation, improve the talent introduction policy, optimize the structure of S&T talent resources, and fully realize the scale effect brought by the clustering of S&T human resources. The provinces that crossed the second threshold of STML are Guangdong and Jiangsu. As China's economic powerhouses, they not only achieved scale in S&T human resources investment, but also ranked at the forefront of coastal provinces in education investment.

- (3) The impact of STMS on EP shows a clear negative double threshold. With the continuous increase of STMS, its negative effect on EP diminishes significantly. Until the first threshold of STMS is crossed, the realization of technological achievements is affected by the level of regional economic development as the investment in science and technology continues to increase. Although the realization capacity of S&T achievements is gradually established, its added value is not easy to grasp. The provinces between the two thresholds account for 36%. These provinces should further improve the management system of technology market, enhance its liquidity, and improve the scale effect of the technology markets. When the second threshold of STMS is crossed, the negative impact of STMS on EP will be further reduced with the continuous improvement of the technology transfer service system. The number of provinces that have crossed this threshold accounts for 55%, including Guangdong, Jiangsu, Shanghai, Shandong, Tianjin, and Liaoning. These provinces should strengthen the construction of high-level professional technology transfer institutions to promote the quantitative to qualitative transformation of technology transactions.

Table 7 Test on threshold effects and estimation of threshold values

Threshold variable	Test on threshold effects		Estimation of threshold values	
	Single-threshold effect test	Double-threshold effect test	Estimated value	95% confidence interval
EFRS	16.64* [0.070]	4.180 [0.740]	1.395	[1.351, 1.444]
STMS	19.66** [0.012]	8.040 [0.260]	0.629	[0.575, 0.650]

The numbers above the square brackets are the F statistics corresponding to the threshold test; ***, **, and * indicate the significance levels at 1%, 5%, and 10%, respectively. The square brackets are the *P* values obtained by repeated sampling using the bootstrap method

- (4) TSC has a significant positive impact on EP, indicating that the continuous accumulation of education manpower investment can strengthen technological progress and achieve economic growth. There is an overlap between education manpower investment and S&T. The output brought about by education investment is the continuous accumulation of human capital, which can deliver high-quality and professional talents for S&T development, realize the specialization of labor, and then promote economic growth (Abel & Gabe, 2011).

The impact of LSS on EP is negative. Blindly expanding the scale of running a school in short term will reduce the ratio of per capita education resources, lower the quality of talent training, and cause damage to the effectiveness of running a school. Localities should reasonably adjust LSS according to the actual input of educational resources, find a balance between scale and efficiency, improve the quality of LSS, and increase economic benefits. RDI has a significant negative impact on EP, probably because of the study period. There are problems such as rigidity, fragmentation, and irregularity in the use of research funds. For example, research funding often faces “fragmentation” problems such as multiple management, duplicate applications for S&T resource allocation, and duplicate investment. In addition, the reimbursement system for S&T research funds is very cumbersome. In fact, only a portion part of S&T research funds is used for research. Corruption can occur if the monitoring system is inadequate. These problems lead to the ineffective use of S&T research funds.

Additional Spatial Analysis

The results of the analysis in the “[Results Analysis](#)” section show that both the S&T innovation environment and educational activities base have significant effects on EP in coastal regions. However, there are significant regional disparities in both S&T innovation and education development in China due to economic, environmental, and policy factors (Du & Seo, 2022). In this way, the empirical results for the coastal regions may not be representative of the less developed regions of China. Therefore, we use the same approach as in the previous section to further explore the impact of S&T innovation and education on EP in central and western regions.

Table 8 Estimation results of model parameters on central regions

Variable	Model 4 Coefficient	Variable	Model 5 Coefficient
TSC	− 0.282 (0.266)	TSC	− 0.296 (0.261)
LSS	1.007*** (0.288)	EFRS	1.192*** (0.273)
STML	0.0227** (0.011)	LSS	0.012* (0.007)
RDI	− 0.074* (0.039)	RDI	− 0.116*** (0.040)
STMS	0.022 (0.013)	STML	0.032*** (0.011)
EFRS_1	0.067*** (0.019)	STMS_1	− 0.101*** (0.033)
EFRS_2	0.009 (0.007)	STMS_2	0.029** (0.013)
Constant	0.146 (0.161)	Constant	0.092 (0.156)
<i>R</i> -squared	0.502	<i>R</i> -squared	0.521

The *t* values corresponding to each coefficient are in parentheses; ***, **, and * indicate the significance level at 1%, 5%, and 10%, respectively

(1) The central regions in China

As seen in Table 7, both educational financial resources support (EFRS) and S&T market scale (STMS) passed the single threshold test at the 1% significance level, indicating the existence of a single threshold for both.

According to Table 8, we found that there is a single threshold characteristic for the impact of EFRS on EP in the central region. When EFRS is less than 13.95 billion yuan, its elasticity coefficient for EP is 0.068. When crossing this threshold, the impact of EFRS is not significant. Compared with the coastal regions, although education resources are generally abundant in central areas, they are unevenly distributed geographically, and high-level universities are scarcer. Therefore, when EFRS crosses the threshold, the asynchronous development of education among the central provinces reduces the efficiency of EFRS utilization. As a result, the economic welfare effect brought by EFRS will be weakened, which in turn will affect the overall economic development of the central regions.

STMS has a significant positive “U” shape impact on EP in the central regions. When STMS exceeds 6.29 billion yuan, its elasticity coefficient for EP changes from − 0.101 to 0.029. The historical experience of innovative development in the central regions is not as rich as that in the coastal areas, and the economic scale that their innovative development base can carry is not as large as that in the coastal regions (Wang et al., 2019). Because of this, with the continuous improvement of

STMS, the economic scale effect brought by the development of the technological market in the central regions will be more easily manifested. Observing the sample data, the provinces crossing the threshold include Hubei, Anhui, Heilongjiang, and Hunan, and there are also significant differences in the STMS levels of these provinces. In practical terms, although the development of the technological market in the central regions brings significant economic effects after a certain scale, such economic impacts are not sustainable. In the long run, there is still a need to narrow the regional gap in innovation development.

In addition, STML has a significant positive impact on EP in the central regions. Observing the sample data, the average annual growth rate of STML in central regions (1.11%) is similar to that in coastal areas (1.12%), which indicates that the rapid growth in the scale of S&T innovation talent provides strong support for the economic rise in central China. The effect of LSS on EP is also significantly positive because the expansion of the scale of schooling can absorb more educational resources, which can enhance local talent cultivation and promote regional economic development. This is different from coastal regions, which focus on high-quality talent cultivation rather than talent scale construction due to policy advantages. RDI has a significant inhibitory effect on EP. Compared with the coastal regions, the average level of RDI in the central areas during the study period is low, which hinders the effective implementation of various innovation activities in the long term, and is not conducive to improving the conversion rate of innovation outputs, thus weakening the dynamics of S&T innovation in economic development.

(2) The western regions in China

As seen in Table 9, both educational financial resources support (EFRS) and S&T manpower level (STML) passed the single threshold test at the 1% significance level, indicating the existence of a single threshold for both.

From Table 10, the impact of EFRS on EP in the western regions is characterized by a significant negative single threshold. There is a large gap between the west and the coast in terms of educational development, which is due to various reasons such as economic and environmental factors and the accumulation of historical factors. The inefficient allocation of educational resources has led to the limited ability of EFRS to serve the economic and social development of the western regions. When EFRS crosses a certain threshold, its inhibiting effect on EP is able to be narrowed. This is because, since the 18th National Congress of the Communist Party of China (CPC), the state has introduced a series of policies and initiatives to support the development of education in the west, which has enhanced the capacity of local educational infrastructure also improved the education environment.

The impact of STML on EP in the west has a significant single threshold feature but the coefficient is not significant, while the impact of STMS on EP shows a clear contribution. From the sample data, the average level of STML in the coastal areas is about 4.5 times higher than that in the west, and the average level of STMS is about 2.8 times higher. In order to promote the development of the western regions, the “Western Development” policy has become one of the most important ways (Zhang

Table 9 Test on threshold effects and estimation of threshold values

Threshold variable	Test on threshold effects		Estimation of threshold values	
	Single-threshold effect test	Double-threshold effect test	Estimated value	95% confidence interval
EFRS	16.640* [0.070]	6.090 [0.506]	4.953	[4.596, 5.220]
STML	13.610* [0.092]	10.760 [0.164]	2.012	[1.789, 2.111]

The numbers above the square brackets are the F statistics corresponding to the threshold test, ***, **, and * indicate the significance levels at 1%, 5%, and 10%, respectively. The square brackets are the P values obtained by repeated sampling using the bootstrap method

Table 10 Estimation results of model parameters on western regions

Variable	Model 6 Coefficient	Variable	Model 7 Coefficient
TSC	− 2.456*** (0.583)	TSC	− 2.601*** (0.573)
LSS	1.507*** (0.474)	LSS	2.210*** (0.418)
STML	0.003 (0.017)	STMS	0.036*** (0.007)
RDI	− 0.002 (0.025)	RDI	0.011 (0.024)
STMS	0.024*** (0.008)	EFRS	− 0.012* (0.006)
EFRS_1	− 0.025*** (0.007)	STML_1	0.009 (0.017)
EFRS_2	− 0.014** (0.006)	STML_2	− 0.015 (0.015)
Constant	0.883*** (0.120)	Constant	0.711*** (0.113)
<i>R</i> -squared	0.530	<i>R</i> -squared	0.542

The *t* values corresponding to each coefficient are in parentheses; ***, **, and * indicate the significance level at 1%, 5%, and 10%, respectively

et al., 2022). With this policy, although the gap between the western region and other regions in terms of technology market development has been narrowed, the lack of talent is still a bottleneck limiting its S & T innovation and economic development. The impact of TSC and LSS on EP in the west is opposed. From the sample data, the west has the lowest level of TSC and LSS compared to other regions. Although the expansion of the schooling scale is conducive to talent cultivation, the lack of faculty can hardly meet the needs of schooling scale expansion, which can make it difficult for regional economic development to obtain a stable flow of talent delivery.

Conclusion

Summary and Conclusions

The article aims to evaluate the effectiveness of the STIEI mechanism and to find the paths to increase the content of technology and knowledge in economic growth. Accordingly, this study first takes the coastal regions, which are the frontier of China's development, as an example. We use DEA tools to process STIEI input–output panel data from 2006 to 2015 to measure and compare the efficiency of the three systems, including STIEI, S&T innovation and education, to verify

the feasibility of the STIEI mechanism in China. Then, considering the reality of regional differences in China, we investigate the impacts of the innovation environment and the foundation of educational activities on economic performance in coastal, central, and western regions separately through panel threshold regression methods. The study found the following results:

1. STIEI mechanism that comprehensively considers S&T innovation and education has relatively higher efficiencies in use of financial, human resources, and other resources.
2. The STIEI mechanism is unable to function at its best and leads to noticeable regional disparities in coastal regions due to the lack of synchronization between S&T innovation and education. With the evolution of STIEI efficiency, a three-legged spatial pattern of “North, Central, and South” emerged at the end of the study period in China’s coastal regions.
3. There is significant regional heterogeneity in the impacts of the innovation environment and the foundation of educational activities on EP. Reasonably strengthening investment in innovative talents and improving the environment of technological market are important for all regions in China. In terms of financial investment in education, the coastal and central regions should improve the utilization, while the western regions should focus on scale enhancement. In addition, coastal regions should focus on improving the quality of schooling, while the central and western regions should reasonably increase the scale of running a school.

Policy Implications

To further enhance the effectiveness of the STIEI mechanism in economic development, it is necessary to start with both S&T innovation and educational subsystems. Based on the conclusions of this paper, there are two policy implications as follows.

First, in terms of education, coastal areas should further strengthen the educational environment and system construction to promote the accumulation of human capital and provide professional talents for S&T innovation (Martinidis et al. 2021). The central regions should further enhance the utilization of educational resources. For western regions that have not yet formed a scale, it is necessary to further appropriately increase investment in education, optimize the space for education development, and based on increased investment, rationally adjust the education financial and human investment structure at different stages of education to reduce investment redundancy. In addition, in terms of the scale of running a school, coastal regions should avoid blindly expanding the scale, and focus on improving the quality of running a school based on the existing educational resource base. And the central and western regions should reasonably increase their scale to promote the realization of the scale effect of education talent training.

Second, in terms of S&T innovation, the coastal areas where financial resources and investment have formed a scale should, on the one hand, further strengthen the construction of leading innovative enterprises, and on the other hand, consider the

actual situation to improve the talent introduction policy and optimize the structure of S&T talent resources. For the central and western regions that have not yet formed a scale, it is necessary to further strengthen S&T innovation investment, improve the efficiency of innovation, and enhance the position of S&T innovation leading economic development. In terms of funding, the inter-ministerial joint conference system for national S&T innovation management should further strengthen, and resources should be integrated and coordinated to avoid invalid in use of funds due to irregular factors. In terms of the technological market, it is necessary to base it on the actual situation of the region. The central and western regions still need to increase the size of the technology market while paying attention to improving laws and regulations on technology transformation. Coastal regions should strengthen the construction of professional technology transformation institutions to enhance the liquidity of the technology market.

Although the article effectively analyzes the influence mechanism of S&T innovation and education on economic development and provides policy recommendations for enhancing the effectiveness of the STIEI mechanism, it lacks analysis and discussion on the cooperative mechanism of inter-regional STIEI. Moreover, the research area and scale need to be combined with the constantly developing knowledge economy to make innovations (especially at the micro level) with important practical value, such as enterprises and universities. In the future, we will improve the limitations and enrich the objectivity of the research.

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Declarations

Conflict of interest The authors declare no competing interests.

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